



DEPARTAMENTO
DE INFORMÁTICA
PUC-RIO

Course Overview



Outline

- **Course organization**
 - Short Biography
 - Learning objectives
 - Bibliography
 - Grading



Alberto Sardinha

Research projects: RELEVANT, HOTSPOT, TAILOR, Ad Hoc Teams, ILU, INSIDE, LIREC

- Multiagent systems: ad hoc teamwork, deep reinforcement learning agents, robotics

Research interests

- Multiagent decision making
- Machine learning
- Real Options
- Application domains: Human-robot interaction, smart mobility, renewable energy

Professional path

- Professor @ DI/PUC-Rio (2023 – present)
- Professor @ IST, Portugal (2010-2023)
- Senior researcher @ INESC-ID, Portugal (since 2012)
- Marie Currie fellow @ Lancaster University, UK (2008 – 2010)
- Post-doc fellow @ Carnegie Mellon University, USA (2005 – 2008)
- System Specialist @ IBM, Brazil (1995 – 1999)



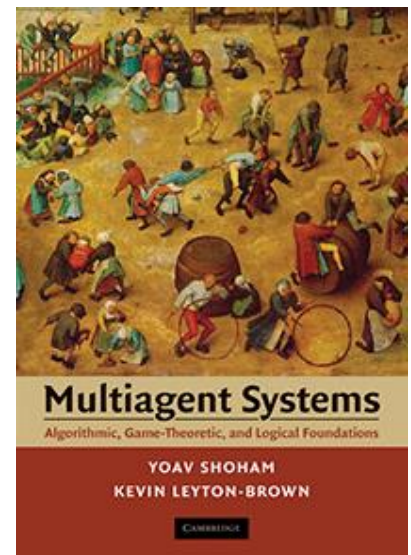
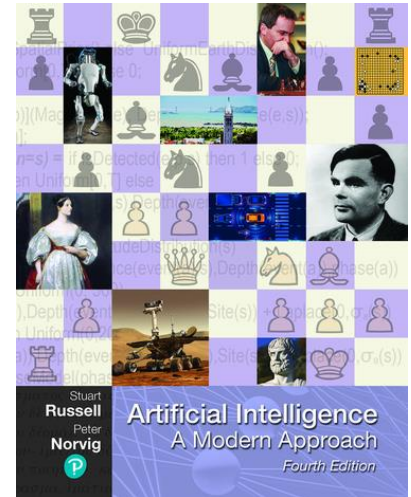
Learning Objectives

- **Models for decision making**
 - Utilities and decision making
 - Sequential decision making under uncertainty
 - Multiagent decision making
- **Learning and decision making**
 - Reinforcement learning
 - Deep reinforcement learning
 - Multiagent learning

Bibliography

Primary

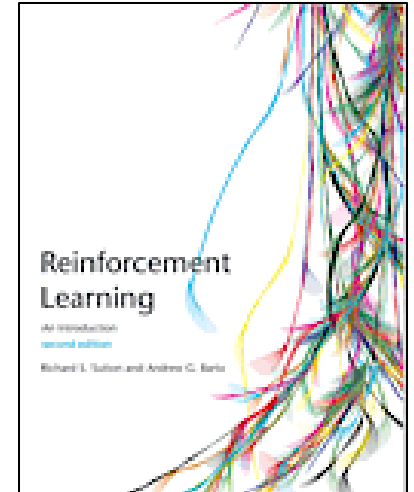
- ***Artificial Intelligence: A Modern Approach***, 4th Ed, Stuart Russel and Peter Norvig, Prentice-Hall, 2021.
- ***Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations***, 1st Ed, Yoav Shoham and Kevin Leyton-Brown, Cambridge University Press, 2009



Bibliography

Primary

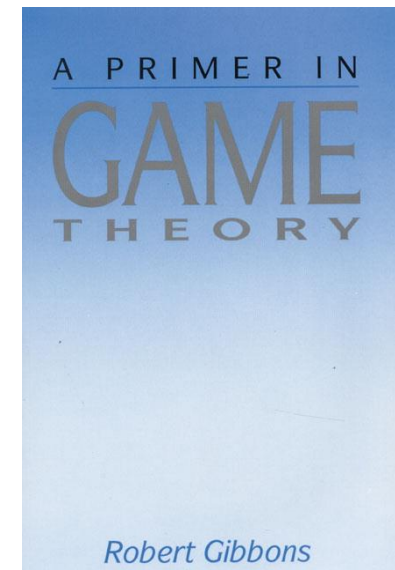
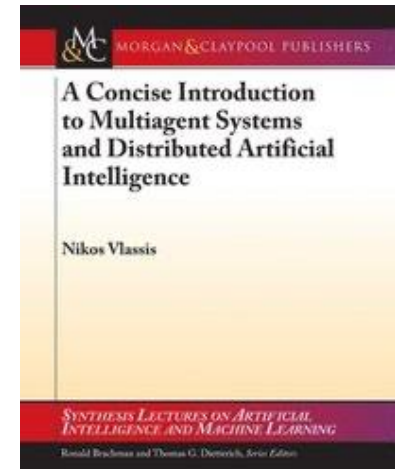
- ***Reinforcement Learning: An Introduction***, 2nd Ed, Richard Sutton, Andrew Barto, MIT Press, 2018.



Bibliography

Secondary

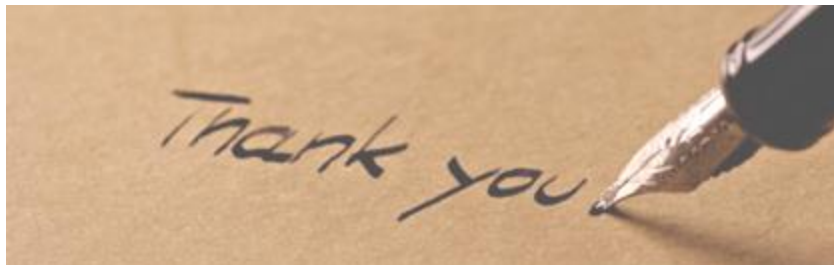
- ***A Concise Introduction to Multiagent Systems and Distributed Artificial Intelligence***, Nikos Vlassis, Morgan & Claypool, 2007
- ***Primer In Game Theory***, Robert Gibbons, Pearson Education Limited, 1992



Grading

- **Project**
 - Environment
 - Implement decision-making algorithms
 - Evaluate algorithms
- **Scientific paper**
- **Presentation**

Thank You



sardinha@inf.puc-rio.br